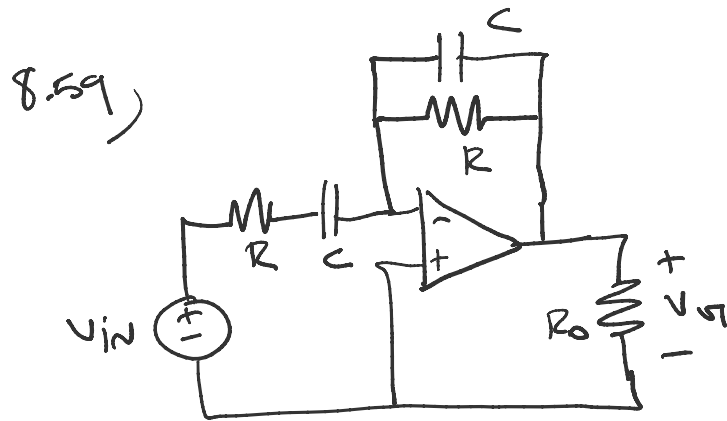


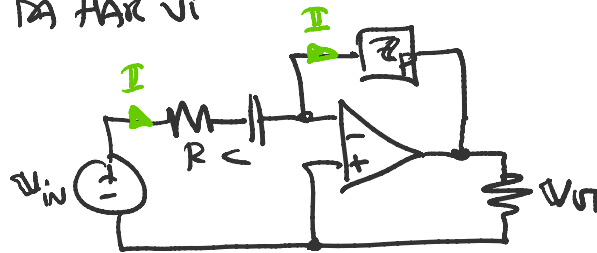


P-kopplingen:

$$\bar{Z}_p^{-1} = \frac{1}{R} + j\omega C = \frac{1 + j\omega CR}{R}$$

$$\Rightarrow \bar{Z}_p = \frac{R}{1 + j\omega CR}$$

Då tar vi



KVL ur

$$-V_{in} + (R + \frac{1}{j\omega C})I = 0 \quad (1)$$

och

$$-V_{out} - \bar{Z}_p I = 0 \Rightarrow I = -\frac{V_{out}}{\bar{Z}_p} \quad (2)$$

(2) i (1)

$$-V_{in} + (R + \frac{1}{j\omega C}) \left(-\frac{V_{out}}{\bar{Z}_p} \right) = 0$$

$$\frac{V_{out}}{V_{in}} = - \frac{\bar{Z}_p}{R + \frac{1}{j\omega C}} = - \frac{\frac{R}{1 + j\omega CR}}{R + \frac{1}{j\omega C}} =$$

$$= - \frac{\frac{j\omega CR}{1 + j\omega CR}}{1 + j\omega CR} = - \frac{j\omega CR}{(1 + j\omega CR)^2} =$$

$$= \frac{j\omega CR}{1 + 2j\omega CR - (\omega CR)^2}$$

$$= \frac{1}{1 + 2j\omega CR - (\omega CR)^2}$$

Tag BeroPPER AN Delta

$$\left| \frac{V_R}{V_{in}} \right| = \frac{\omega CR}{\sqrt{(1 - (\omega CR)^2)^2 + (2\omega CR)^2}} \quad (*)$$

$$= \frac{\omega CR}{(1 - 2(\omega CR)^2 + (\omega CR)^4 + 4(\omega CR)^2)^{1/2}}$$

$$= \frac{\omega CR}{\sqrt{(1 + (\omega CR)^2)^2}}$$

$$= \frac{\omega CR}{1 + (\omega CR)^2}$$

Altså

$$|H(\omega)| = \frac{\omega CR}{1 + (\omega CR)^2} \quad (**)$$

Resonansfrekvens kan man bestämma
UR (*) OCH DA NÄMNER. ÄR EN MINST,
DVS DA

$$1 - (\omega_0 CR)^2 = 0$$

$$\omega_0 = \frac{1}{CR}$$

ELLER UR DERIVATAN AV (**). Se

SPRISATI Lsg i CANVAS, 8.59-2. Rlf